

2045	0.70	0.60	0.45	0.46	0.46	0.50	0.58	0.71	0.80	0.86
2050	0.67	0.56	0.39	0.40	0.40	0.44	0.53	0.66	0.77	0.87
2055	0.64	0.52	0.34	0.35	0.35	0.39	0.48	0.62	0.73	0.88
2060	0.61	0.48	0.30	0.30	0.31	0.34	0.43	0.58	0.70	0.87

D.1.3 Research Incidence Data

Several additional incidence datasets are included to support stratified mortality data. Some recent epidemiological studies include concentration-response functions which are stratified by subgroups. This section describes additional datasets which are included to support subgroup stratification.

D.1.3.1 Ethnicity-Stratified Mortality Incidence

To estimate ethnicity-stratified and age-stratified incidence rates at the county level, we downloaded all-cause and respiratory mortality data from 2007 to 2016 from the CDC WONDER mortality database (<https://wonder.cdc.gov/>). Ethnicity-stratified incidence rates were calculated for the following age groups: < 1 year, 1-4 years, 5-14 years, 15-24 years, 25-34 years, 35-44 years, 45-54 years, 55-64 years, 65-74 years, 75-84 years, and 85+ years. We stratified county-level data by two groups, Hispanic and non-Hispanic, and did not stratify further by race due to suppression constraints.

We followed the methods outlined in section D.1.1 to deal with suppressed and unreliable data, with one notable difference in methodology; we included an intermediate spatial scale between county and state for imputation purposes. We designated urban and rural counties within each state using CDC WONDER and, where possible, imputed missing data using the state-urban and state-rural classifications before relying on broader statewide data. We followed methods for dealing with suppressed and unreliable data at each spatial scale as described in section D.1.1.

D.1.3.2 Ethnicity-Stratified Mortality Projections (2020-2060)

Baseline ethnicity-stratified mortality rates were derived from county-level CDC WONDER data for 2007-2016 (see section D.1.3). In order to project these rates into future years, we procured national-level mortality projections from the U.S. Census Bureau. These projections cover the years 2017 to 2060. Because county-level mortality projections are not readily available, we apply the national rates of change to the baseline county values. In addition, we bridge the temporal gap between the 2007-2016 period—treated as a 2011 average—and the 2017-onwards projection using historical CDC WONDER data for 2011 to 2017.

Table D-5. Datasets Used in Developing Ethnicity-Stratified Incidence Rates (2020-2060)

Initial Baseline Dataset	CDC WONDER Cause of Death Dataset - for adjustment to 2017	Census Bureau Population Projection	Census Bureau Deaths Projections
Years: 2007-2016 Ages: < 1 year, 1-4 years, 5-14 years, 15-24 years, 25-34 years, 35-44 years, 45-54 years, 55-64 years, 65-74 years, 75-84 years, and 85+ years Spatial scale: County-level	Years: 2011, 2012, 2017 Ages: < 1 year, 1-4 years, 5-14 years, 15-24 years, 25-34 years, 35-44 years, 45-54 years, 55-64 years, 65-74 years, 75-84 years, and 85+ years Spatial scale: Census regions	Years: 2017-2060 Ages: 0, 1, 2,...99, 100 Spatial scale: National	Years: 2017-2060 Ages: 0, 1, 2, ...84, 85+ Spatial scale: National

D.1.3.3 Adjusting Baseline Dataset to 2017 Using CDC WONDER Historical Data

First, the baseline dataset was adjusted to 2017 values using historical data from CDC WONDER's Underlying Cause of Death dataset (<https://wonder.cdc.gov/ucd-icd10.html>). All-cause mortality datasets were pulled for years 2011, 2012, and 2017, for Hispanic and Non-Hispanic groups. Data was aggregated from the Census Region to a national level. Values for years 2011 and 2012 were averaged together to represent the midpoint of the 2007 to 2016 baseline rates. A scaling factor was calculated between the 2011-2012 average and 2017 values. These scaling factors (Table D-6) were applied to the original, county-level baseline mortality rates to adjust the dataset to represent 2017 mortality incidence.

Table D-6. Ratio of 2017 to 2011-2012 All-Cause Mortality Rate

Start Age	End Age	Hispanic	Non-Hispanic
0	0	1.003709	0.933217
1	4	0.857753	0.941917
5	14	1.006885	1.067448
15	24	1.141657	1.101791
25	34	1.282166	1.261146
35	44	1.162141	1.147084
45	54	0.988912	0.998058
55	64	0.987971	1.049693
65	74	0.970447	0.983404
75	84	0.947003	0.951082

85	99	0.939672	0.996925
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D.1.3.4 Projecting Dataset into 2060 Using Census Bureau Projections

To project this adjusted 2017 baseline dataset into the future, we used a similar process to the methods described in Section D.1.2. We obtained Census Bureau projected deaths and population for 2017-2060 at the national level and adjusted the age groups to match between datasets. The Census Bureau datasets were stratified by ethnicity (Hispanic and Non-Hispanic). We calculated all-cause mortality rates by dividing projected deaths by projected population.

We then calculated, separately for each age group and ethnicity group combination, the ratio of the Census Bureau national mortality rate in year Y (Y= 2017, 2018, ...2060) to the 2017 rate. These ratios are shown for a selection of years and age groups in Table D-7.

Table D-7. Ratio of Future All-Cause Mortality Rate to 2017 Estimated All-Cause Mortality Rate

Ethnicity	Year	Age Group					
		0-0	5-14	25-34	45-54	65-74	85-99
Hispanic	2020	0.98	0.97	0.94	0.94	0.97	0.99
	2030	0.93	0.89	0.76	0.82	0.88	0.92
	2040	0.87	0.79	0.61	0.70	0.80	0.87
	2050	0.82	0.71	0.48	0.59	0.73	0.86
	2060	0.77	0.63	0.38	0.49	0.66	0.86
Non-Hispanic	2020	0.97	0.96	0.93	0.93	0.97	1.00
	2030	0.87	0.82	0.69	0.71	0.86	0.91
	2040	0.79	0.72	0.52	0.57	0.75	0.86
	2050	0.72	0.62	0.40	0.45	0.63	0.86
	2060	0.65	0.54	0.30	0.36	0.56	0.87

Finally, the ratios displayed in Table D-7 were used to develop projected mortality datasets on the county level stratified by ethnicity (Hispanic and Non-Hispanic) for years 2020, 2025, ... 2055, 2060. We multiplied each county's incidence value with the scaling ratio corresponding to its specific age group and ethnicity for years 2020-2060.

D.1.3.5 Race-Stratified Mortality Incidence

To estimate race-stratified and age-stratified incidence rates at the county level, we downloaded all-cause and respiratory mortality data from 2007 to 2016 from the CDC WONDER mortality database (<https://wonder.cdc.gov/>). Race-stratified incidence rates were calculated for the following age groups: < 1 year, 1-4 years, 5-14 years, 15-24 years, 25-34 years, 35-44 years, 45-54 years, 55-64 years, 65-74 years, 75-84 years, and

85+ years.⁴⁰ To address the frequent county-level data suppression for race-specific death counts, we stratified the county-level data into two broad race categories, White and Non-White. In a later step, we stratified the non-White incidence rates by race (Black, Asian, Native American) using the relative magnitudes of incidence values by race at the regional level, described in more detail below. We also included an intermediate spatial scale between county and state designating urban and rural counties for imputation purposes, described in detail in section D.1.1.

A pooled non-White incidence rate inherently underestimates the mortality risk for some race groups and overestimates mortality risk for others. To estimate county-level mortality rates by individual race (Black, Asian, Native American), we applied regional race-specific incidence relationships to the county-level pooled non-White incidence rates. We calculated a weighted average of race-specific incidence rates using regional incidence rates for each region/age/race group normalized to one reference population (the Asian race group) and county population proportions based on race-specific county populations from CDC WONDER where available. In cases of population suppression across two or more races per county, we replaced all three race-specific population proportions derived from CDC WONDER with population proportions derived from 2020 Census data in BenMAP-CE.

D.1.3.6 Race-Stratified Mortality Projections (2020-2060)

The methods for projecting race-stratified mortality incidence rates are comparable to those described in section D.1.4 for ethnicity-stratified rates. The only difference includes mapping different definitions of race across datasets, as summarized in Table D-8. Due to existing population configurations in BenMAP-CE, the mortality projections were adjusted to match the limited set of racial groups currently in the tool.

⁴⁰ Infant mortality dates for race- and ethnicity-stratified datasets do not currently exclude neonatal deaths.